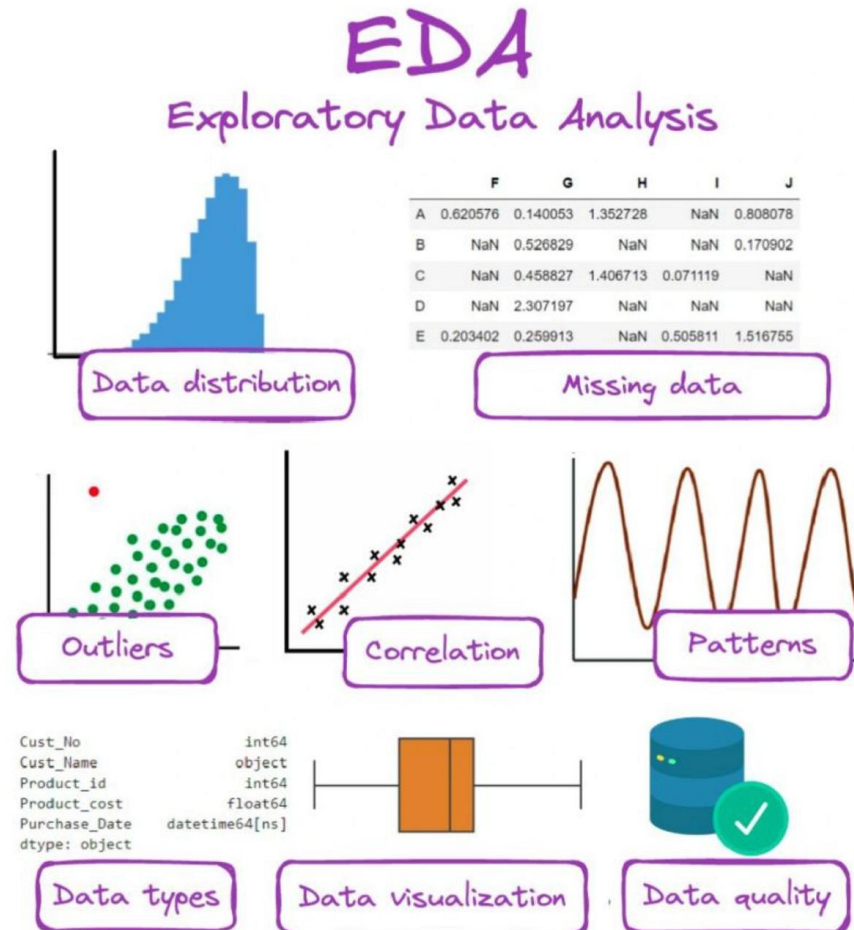
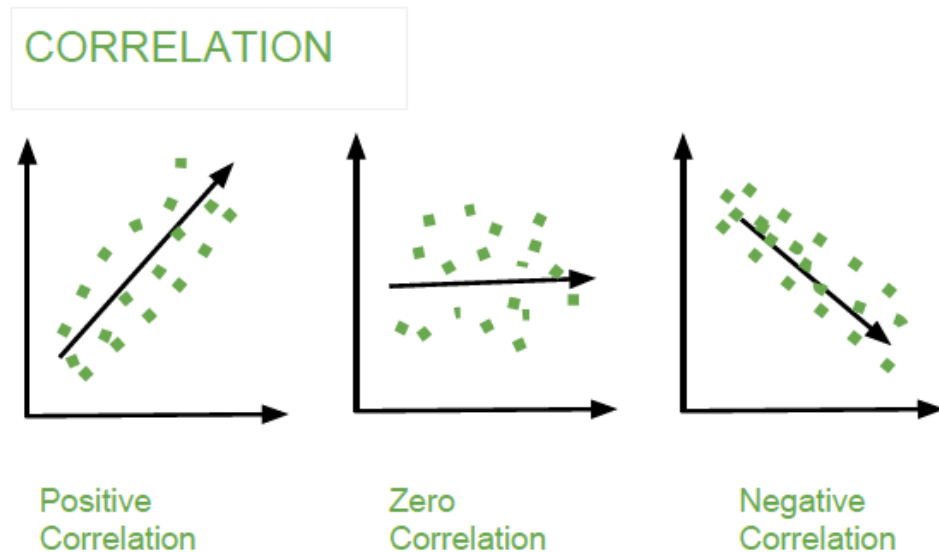


The Shape of Data (Beyond the Mean)

- **Moving Beyond Averages:** The mean and median only tell part of the story. To truly understand data, we look at its "Physique."
- **Skewness:** Does the data lean to the left or right? (e.g., Income data usually has a "Long Tail" to the right).
- **Kurtosis:** Is the data "Pointy" (heavy outliers) or "Flat" (spread out)?
- **Why it matters:** Many AI algorithms assume a perfect bell curve. If your data is too skewed, the model will fail.



Correlation – Finding Hidden Connections



- **Definition:** A mathematical value between -1 and 1 that shows how two variables move together.
- **Heatmaps:** A visual grid where colors represent the strength of a relationship.
- **Positive vs. Negative:**
 - **Positive:** As Education increases, Income tends to increase.
 - **Negative:** As Car Weight increases, Fuel Efficiency decreases.
- **Warning:** Correlation does **not** equal causation!

Bivariate & Multivariate Analysis

- **Bivariate:** Comparing two variables (e.g., Rainfall vs. Crop Yield). We use Scatter Plots to see the relationship.
- **Multivariate:** Watching 3 or more variables interact simultaneously.
- **The "Hue" Dimension:** Adding color to a scatter plot to represent a third category (e.g., Comparing Rainfall vs. Yield, colored by "Province").
- **Pair Plots:** A "Grand Tour" of the dataset where every variable is plotted against every other variable.

